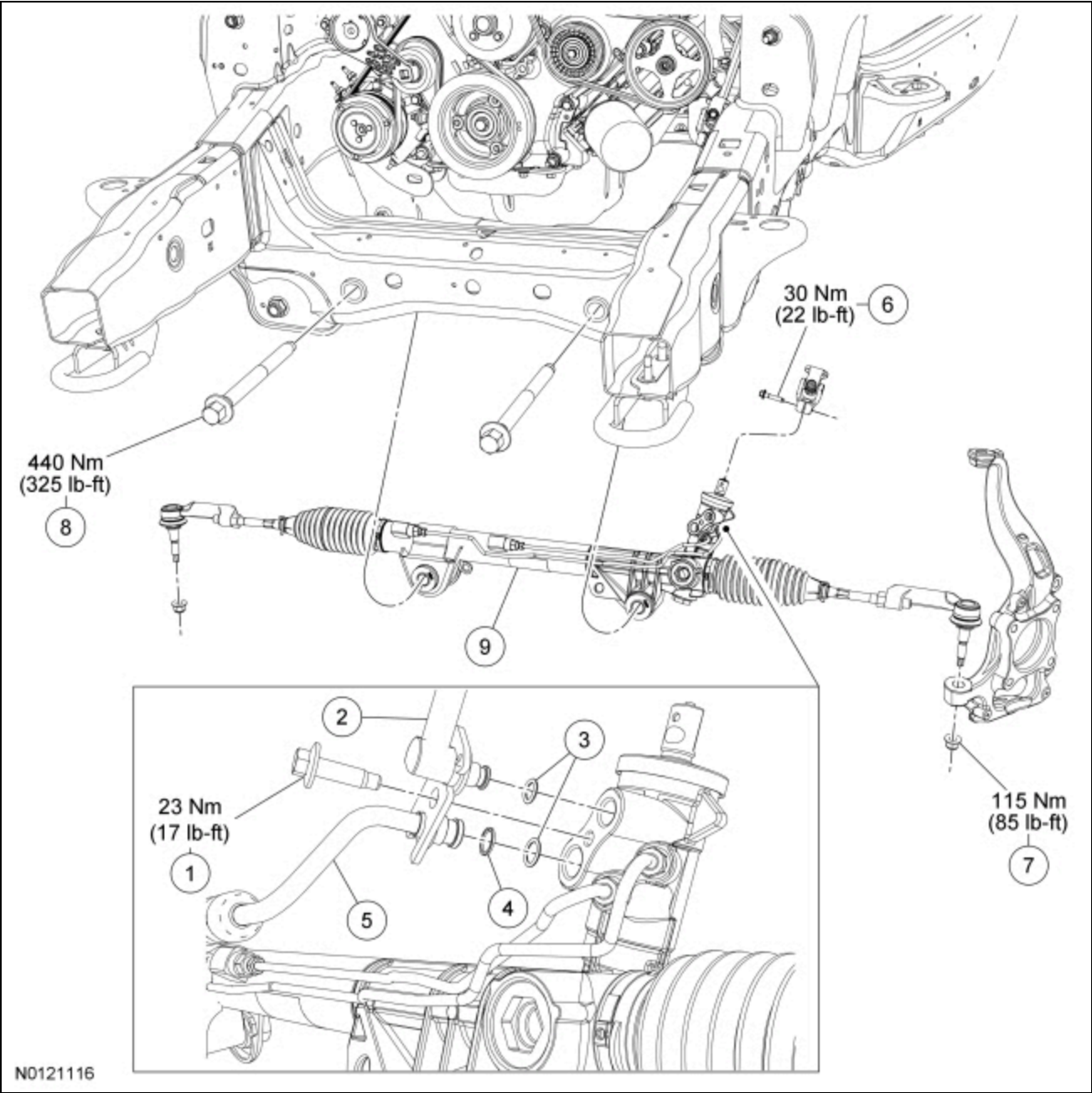


Steering Gear — Hydraulic Power Assist Steering (HPAS)

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)	
 ST2945-A	Separator, Ball Joint Tool 204-592



Item	Part Number	Description
1	W712679 W712679	Steering line clamp plate-to-steering gear bolt

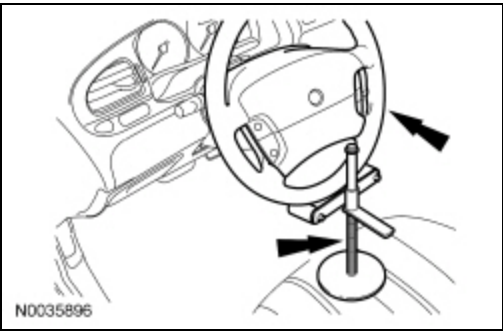
2	3A713C 3A713C	Return line
3	3F886 3F886	O-ring seals (2 required)
4	3B602 3B602	Back-up O-ring seal
5	3A719 3A719	Pressure line
6	N808684 N808684	Steering column shaft-to-steering gear bolt
7	W520215 W520215	Outer tie-rod end nut (2 required)
8	W713954 W713954	Steering gear-to-crossmember bolt (2 required)
9	3504 3504	Steering gear

Removal and Installation

NOTICE: While repairing the power steering system, care should be taken to prevent the entry of foreign material or failure of the power steering components may result.

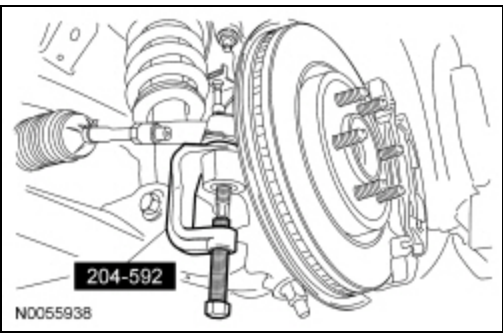
1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. **NOTICE:** Do not allow the steering column shaft to rotate while disconnected from the gear or damage to the clockspring may occur. If there is evidence that the steering column shaft has rotated, remove and recenter the clockspring. For additional information, refer to Section 501-20B.

Use a suitable holding device and hold the steering wheel in the straight-ahead position.



3. Remove the front wheels and tires. For additional information, refer to Section 204-04.
4. If equipped, remove the skid plate bolts and the skid plate.
 - To install (all except SVT Raptor), tighten to 25 Nm (18 lb-ft).
 - To install (SVT Raptor), tighten to 48 Nm (35 lb-ft).
5. Remove and discard the outer tie-rod end nuts.
 - To install, tighten to 115 Nm (85 lb-ft).
6. **NOTICE:** Use care when installing the Ball Joint Tool Separator or damage to the tie-rod end boot may occur.

Using the Ball Joint Tool Separator, disconnect the tie-rod ends from the wheel knuckles.



7. **NOTICE: Do not allow the steering column shaft to rotate while disconnected from the gear or damage to the clockspring may occur. If there is evidence that the steering column shaft has rotated, remove and recenter the clockspring. For additional information, refer to Section 501-20B.**

Remove and discard the steering column shaft-to-steering gear bolt and disconnect the steering column shaft from the steering gear.

- To install, tighten to 30 Nm (22 lb-ft).

8. **NOTICE: Install new O-ring seals any time the lines are disconnected from the steering gear or a fluid leak may occur.**

Remove the steering line clamp plate-to-steering gear bolt, and disconnect the power steering lines.

- Discard the O-ring seals.
- To install, tighten the steering line clamp plate-to-steering gear bolt to 23 Nm (17 lb-ft).

9. Remove and discard the 2 steering gear-to-crossmember bolts and remove the steering gear.

- To install, tighten to 440 Nm (325 lb-ft).

10. **NOTICE: Make sure the tie-rod end boots are seated correctly on the tie-rod ends or tie-rod end failure may occur.**

To install, reverse the removal procedure.

11. Fill the power steering system. For additional information, refer to Section 211-00B.

12. Check and, if necessary, adjust the front toe. For additional information, refer to Section 204-00.